

1(a)(i)	micrometer (screw gauge)/digital calipers	B1
1(a)(ii)	take several readings (and average)	M1
	along the wire or around the circumference	A1
1(b)(i)	$\sigma = 4 \times 25 / [\pi \times (0.40 \times 10^{-3})^2] = 1.99 \times 10^8 \text{ N m}^{-2}$ or $\sigma = 25 / [\pi \times (0.20 \times 10^{-3})^2] = 1.99 \times 10^8 \text{ N m}^{-2}$	A1
1(b)(ii)	$\%F = 2\%$ and $\%d = 5\%$ or $\Delta F / F = \frac{0.5}{25}$ and $\Delta d / d = \frac{0.02}{0.4}$	C1
	$\%\sigma = 2\% + (2 \times 5\%)$ or $\%\sigma = [0.02 + (2 \times 0.05)] \times 100$ $\%\sigma = 12\%$	A1
1(b)(iii)	absolute uncertainty $= (12 / 100) \times 1.99 \times 10^8$ $= 2.4 \times 10^7$	C1
	$\sigma = 2.0 \times 10^8 \pm 0.2 \times 10^8 \text{ N m}^{-2}$ or $2.0 \pm 0.2 \times 10^8 \text{ N m}^{-2}$	A1